```
clear; clc; close all;

Fs = 10000;
fp = 1500;
fs = 2000;
N = 40;

% 1. Parks-McClellan (Equiripple) Design
f_bands = [0 fp fs Fs/2] / (Fs/2);
a_bands = [1 1 0 0];

b_pm = firpm(N, f_bands, a_bands);

% 2. Frequency-Sampling Design
b_fs = fir2(N, f_bands, a_bands);

% Display Numerical Outputs
fprintf('\n=== EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY SAMPLING ===\n\n');
```

```
=== EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY SAMPLING ===
```

```
fprintf('Filter Order (N): %d\n', N);
```

```
Filter Order (N): 40
```

```
fprintf('Passband Edge: %d Hz, Stopband Edge: %d Hz\n\n', fp, fs);
```

```
Passband Edge: 1500 Hz, Stopband Edge: 2000 Hz
```

```

fprintf('--- PARKS-MCCLELLAN COEFFICIENTS (First 10) ---\n');
```

```
--- PARKS-MCCLELLAN COEFFICIENTS (First 10) ---
```

```
disp(b_pm(1:10));
```

```
   -0.0015    0.0072    0.0053   -0.0010   -0.0081   -0.0072    0.0037    0.0142    0.0100   -0.0090
```

```

fprintf('--- FREQUENCY-SAMPLING COEFFICIENTS (First 10) ---\n');
```

```
--- FREQUENCY-SAMPLING COEFFICIENTS (First 10) ---
```

```
disp(b_fs(1:10));
```

```
    0.0000    0.0001    0.0002   -0.0001   -0.0008   -0.0009    0.0008    0.0035    0.0029   -0.0039
```

```

% Frequency Responses
[H_pm, f] = freqz(b_pm, 1, 1024, Fs);
[H_fs, ~] = freqz(b_fs, 1, 1024, Fs);

% Plot Magnitude Responses
figure('Name', 'Parks-McClellan vs Frequency-Sampling');
```

```
subplot(2,1,1);

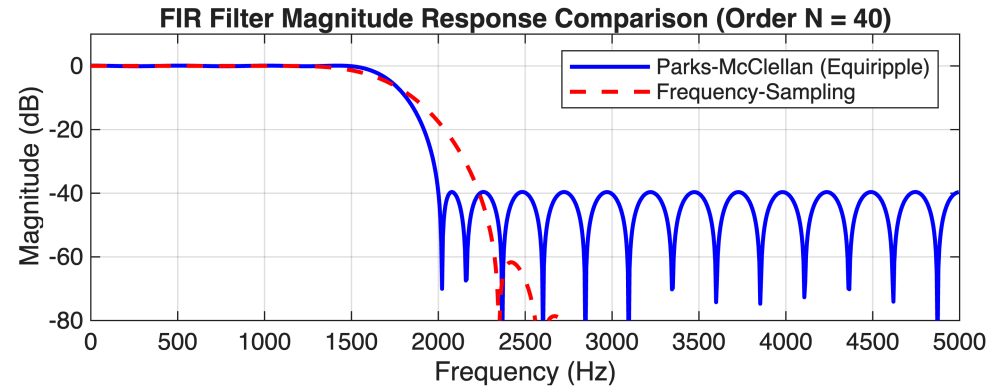
plot(f, 20*log10(max(abs(H_pm),1e-10)), 'b', 'LineWidth', 1.5);
hold on;
plot(f, 20*log10(max(abs(H_fs),1e-10)), 'r--', 'LineWidth', 1.5);

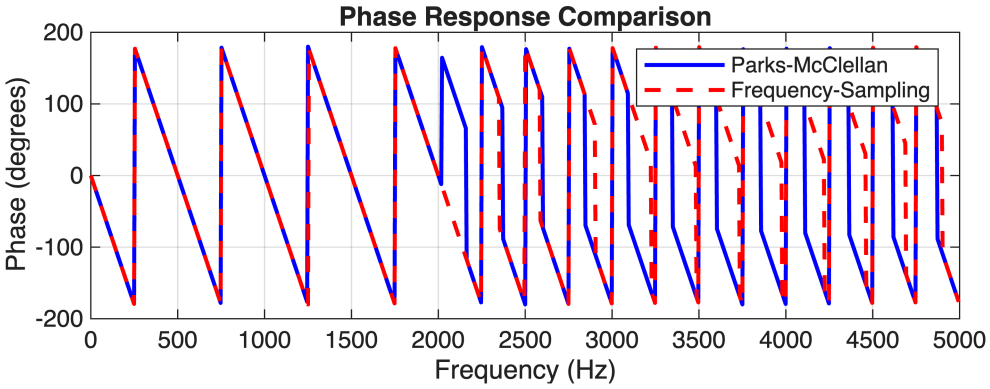
grid on;
legend('Parks-McClellan (Equiripple)', 'Frequency-Sampling');
title(['FIR Filter Magnitude Response Comparison (Order N = ', num2str(N),
')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-80 10]);

% Plot Phase Responses
subplot(2,1,2);

plot(f, angle(H_pm)*180/pi, 'b', 'LineWidth', 1.5);
hold on;
plot(f, angle(H_fs)*180/pi, 'r--', 'LineWidth', 1.5);

grid on;
legend('Parks-McClellan', 'Frequency-Sampling');
title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```


FIR Filter Magnitude Response Comparison (Order N = 40)
Parks-McClellan (Equiripple)
Frequency-Sampling
Magnitude (dB)
0
-20
-40
-60
-80
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
Frequency (Hz)


Phase Response Comparison
Parks-McClellan
Frequency-Sampling
Phase (degrees)
200
100
0
-100
-200
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
Frequency (Hz)